

Impact Statement

Project Title: Domain-Specific Small Language Model for African Folktales

Team Name: Ethopie

Course Track: Google DeepMind: AI Research Foundations

1. Purpose of the Project

The primary purpose of this project is to build a domain-specific Small Language Model (SLM) engineered explicitly for the preservation and creative generation of African folktales. Traditional preservation models struggle because regional histories rely heavily on manual transcription, scattered anthropological records, and vulnerable oral paths. Furthermore, mainstream Large Language Models (LLMs) are predominantly pretrained on massive Western text corpora. Consequently, they fail to generate authentic African narratives, frequently misrepresenting cultural nuances, local dialects, and core oral structures. Team Ethopie's project directly targets this systemic gap by training a lightweight, resource-efficient model on curated oral history translations, published regional literature, and authentic mythologies.

2. Positive Impacts

This SLM offers substantial benefits across social, educational, and technological sectors. Culturally, it provides local communities and global archives with an interactive tool to revitalize storytelling traditions, keeping ancestral wisdom accessible to younger, digitally native generations. Educationally, schools can leverage the model to generate accurate, dialect-aware reading materials that reinforce indigenous literacy. Technologically, this project serves as a key proof of concept demonstrating that optimized, small-scale AI models can achieve high linguistic performance without requiring massive, cost-prohibitive computational infrastructure, thereby empowering marginalized developer communities.

3. Potential Risks and Harms

A major cultural risk involves the accidental homogenization or distortion of sacred localized motifs, as merging diverse storytelling traditions into a single dataset could erode the distinct identities of unique tribal groups. Additionally, sourcing translated anthropological texts presents an economic and ethical risk of data exploitation if local communities are disconnected from the ownership of their digitized heritage. There is also an unintended social consequence where synthetic story generation might inadvertently lower the perceived value of real-world oral historians.

4. Values, Mitigation Strategies, and Limitations

Our framework prioritizes **Inclusivity, Dignity, and Transparency**. To support inclusivity and preserve distinct cultural identities, our dataset explicitly structures and tags stories by their exact historical region and dialect group. To uphold dignity and respect intellectual autonomy, Team Ethopie implements strict ethical data filtration guidelines. We actively avoid sacred or closed narratives that community elders have explicitly requested remain untranscribed, and prioritize public, historically verified anthropological works.

An inherent uncertainty exists regarding the historical accuracy of early colonial text translations within our dataset. Furthermore, a technical limitation is that a small language model may occasionally generate narrative hallucinations or exhibit repetitive text patterns under longer context lengths. We plan to address these limitations through continuous human-in-the-loop evaluation and iterative hyperparameter tuning.